

Shell, Unix, & VIM — MCQ

One correct option per question. No negative marking.

1. In a typical Unix prompt, the symbol for the superuser (root) is:
 - A. \$
 - B. #
 - C. %
 - D. ~

Solution: By convention, normal users see \$; root sees #. This helps you notice when you have elevated privileges.

2. Which command prints the current working directory?
 - A. **pwd**
 - B. **cd**
 - C. **ls**
 - D. **whoami**

Solution: **pwd** expands to the absolute path of the current directory; **cd** changes it.

3. Which option makes **ls** show hidden files?
 - A. **-l**
 - B. **-h**
 - C. **-a**
 - D. **-R**

Solution: Files starting with **.** are hidden; **ls -a** includes them. **-l** is long listing, **-R** is recursive.

4. In the leftmost character of **ls -l** output, a leading **d** indicates:
 - A. A symbolic link
 - B. A directory
 - C. A device file
 - D. An executable file

Solution: The first character is the file type: **d**=directory, **-**=regular file, **l**=symlink, **c/b**=device.

5. For the mode string **-rwxr-xr--**, members of the group can:
- A. Read only
 - B. Write only
 - C. Read and execute
 - D. Read and write

Solution: Owner=**rwx** (7), group=**r-x** (5), others=**r--** (4). Group has read+execute.

6. Which octal mode sets **rw-r--r--**?
- A. **644**
 - B. **755**
 - C. **600**
 - D. **700**

Solution: Owner **rw--**=6, group **r--**=4, others **r--**=4 $\Rightarrow$ **644**.

7. Which redirection sends only standard error to a file?
- A. **>**
 - B. **>>**
 - C. **2>**
 - D. **1>**

Solution: FD 2 is **stderr**. **2>** redirects errors; **1>** redirects **stdout**; bare **>** is equivalent to **1>**.

8. The file descriptor number for **stdout** is:
- A. **0**
 - B. **1**
 - C. **2**
 - D. **3**

Solution: FDs: 0=**stdin**, 1=**stdout**, 2=**stderr**.

9. Recursively and case-insensitively search for “config” under **/etc**:

- A. **grep -r "config" /etc**
- B. **grep -ri "config" /etc**
- C. **find /etc -name "config"**
- D. **grep -R /etc "config"**

Solution: **-r** is recursive; **-i** ignores case. The last option misorders arguments.

10. To view the end of a log file and follow new lines in real time, use:

- A. **head -f file.log**
- B. **tail -f file.log**
- C. **less -f file.log**
- D. **cat -f file.log**

Solution: **tail -f** follows appended lines. (Tip: **tail -F** handles log rotation.)

11. Which command opens the manual page for **ls**?

- A. **help ls**
- B. **ls --man**
- C. **man ls**
- D. **info ls**

Solution: **man ls** shows the manual; **help** is for shell built-ins; **info** is Texinfo (works but not the Unix “man page” request).

12. Set permissions to **rwxr-x---** for **script.sh**:

- A. **chmod 755 script.sh**
- B. **chmod 750 script.sh**
- C. **chmod u=rw,g=rx,o=rx script.sh**
- D. **chmod 700 script.sh**

Solution: **rw****x**=7 for owner, **r-x**=5 for group, and no permissions=0 for others $\Rightarrow$ **750**. The symbolic form shown grants others read+execute, which is incorrect here.

13. What is the default mode when you open Vim?

- A. Insert mode
- B. Visual mode
- C. Command-line mode
- D. Normal mode

Solution: Vim starts in Normal mode, where you navigate and issue commands. From there you switch to other modes like Insert or Visual.

14. Which key is used to enter Insert mode and start typing before the cursor?

- A. **i**
- B. **a**
- C. **o**
- D. **I**

Solution: **i** inserts before the cursor; **a** inserts after, **o** opens a new line below, and **I** inserts at the start of the line.

15. How do you exit Insert mode and return to Normal mode?

- A. **Ctrl+C**
- B. **Alt+X**
- C. **Esc**
- D. **Tab**

Solution: **Esc** cleanly leaves Insert mode. While **Ctrl+C** can interrupt, it isn't equivalent to **Esc** (e.g., it may skip some autocommands).

16. Which command is used to save a file in Vim?

- A. **:s**
- B. **:w**

- C. **:save**
- D. **:write**

Solution: **:w** writes the current buffer to disk. (You can also use **:write**, but the canonical short form is **:w**.)

17. What does the motion command **w** do in Normal mode?
- A. Move to the beginning of the current word
 - B. Move to the beginning of the next word
 - C. Move to the end of the current word
 - D. Move up one line

Solution: **w** jumps to the start of the next word. Related: **e** to the end of the current/next word, **b** to the start of the previous word.

18. Which command deletes the current line in Normal mode?
- A. **d**
 - B. **d␣**
 - C. **dd**
 - D. **dw**

Solution: **dd** is “delete line.” **d** needs a motion (e.g., **dw** delete word, **d␣** delete one char).

19. What does the command **:1,20>>** do?
- A. Delete lines 1 through 20
 - B. Copy lines 1 through 20
 - C. Shift lines 1 through 20 to the right by one **shiftwidth**
 - D. Move lines 1 through 20 to the end of file

Solution: **>>** indents by one **shiftwidth**. With a range **:1,20**, it indents those lines.

20. In Visual mode, which key combination selects text in block/column mode?
- A. **v**

- B. **V**
- C. **Ctrl+v**
- D. **Shift+v**

Solution: **Ctrl+v** (often shown as **^v**) enters Visual block mode; **v** is character-wise, **V** is line-wise.

21. What does the command **ci"** do in Normal mode?
- A. Copy text inside double quotes
 - B. Change (delete and enter insert mode) text inside double quotes
 - C. Cut text inside double quotes
 - D. Count text inside double quotes

Solution: **c** = change (delete + go to Insert). **i"** means "inside quotes"; **ci"** replaces the quoted text and puts you in Insert mode.

22. Which substitution command replaces all occurrences of **foo** with **bar** in the entire file, asking for confirmation each time?
- A. **:%s/foo/bar/g**
 - B. **:%s/foo/bar/c**
 - C. **:%s/foo/bar/gc**
 - D. **:%s/foo/bar/gi**

Solution: **%** targets the whole file. **g** = all matches per line; **c** = confirm each replacement. **i** would add case-insensitive matching.

23. What does the command **:r !ls** do?
- A. Run the **ls** command and display output temporarily
 - B. Replace current line with **ls** command output
 - C. Read (insert) the output of **ls** into the buffer
 - D. List all files in the current directory

Solution: **:r !cmd** runs a shell command and reads its output into the buffer below the current line.

24. If you want to yank (copy) a line into named register **a**, which command would you use?

- A. **yya**
- B. **y"a**
- C. **"a yy**
- D. **a yy**

Solution: **"a** selects register **a**; **yy** yanks the whole line. So **"a yy**. (Alternatives: **"a Y**.)

25. What does the command **3dw** do in Normal mode?

- A. Delete 3 characters
- B. Delete the 3rd word
- C. Delete 3 words starting from cursor position
- D. Delete 3 lines

Solution: A count prefixes the operator: **3dw** = delete (**d**) three word motions (**w**) from the cursor forward.

26. In the substitution command **:%s/^/# /**, what does the **^** pattern match?

- A. Any character at the end of line
- B. Empty lines only
- C. The beginning of each line
- D. Lines containing only spaces

Solution: In Vim regex, **^** anchors the start of a line. The command prefixes each line with **#** .

27. When using the confirmation flag **c** in substitution, what does pressing **a** do?

- A. Abort the substitution completely
- B. Apply substitution to current match only
- C. Replace this and all remaining matches without further prompts
- D. Ask again for the next match

Solution: During **/gc** confirmation, keys include: **y**=yes, **n**=no, **a**=all remaining, **q**=quit, **l**=last, **Ê/Ŷ**=scroll.